

BOROD, RomaniaWMO: **150950**Lat: **46.9936N**Lon: **22.5903E**Elev: **334**StdP: **97.37**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.6	-9.2	-14.8	1.1	-10.4	-12.4	1.3	-8.0	9.7	3.3	8.1	3.3	0.9	120	0.462

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.9	31.3	20.9	29.7	20.4	27.9	19.8	22.1	29.0	21.3	28.0	20.6	26.9	2.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.9	15.2	25.3	19.0	14.4	24.2	18.2	13.7	23.6	66.2	29.3	63.3	28.1	60.6	26.9	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.8	6.3	5.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	-0.9	2.4	5.2	10.4	15.3	18.9	19.9	20.5	15.8	10.9	6.9	1.1
	DBStd	8.35	4.88	4.45	4.24	4.69	3.38	3.20	2.91	3.11	4.30	4.18	4.44	4.47
	HDD10.0	1201	339	215	157	53	4	0	0	0	3	41	112	276
	HDD18.3	3063	597	447	408	239	104	31	18	14	98	231	342	533
	CDD10.0	1408	0	1	7	63	167	268	307	326	178	69	21	1
	CDD18.3	230	0	0	0	1	9	49	66	82	22	1	0	0
	CDH23.3	2270	0	0	0	17	100	472	617	836	210	17	0	0
	CDH26.7	678	0	0	0	1	10	116	182	317	52	0	0	0
Wind	WSAvg	1.7	1.4	1.7	2.1	2.1	1.8	1.6	1.7	1.4	1.5	1.5	1.5	1.5
Precipitation	PrecAvg	982	57	52	62	73	96	123	120	82	91	78	71	77
	PrecMax	1320	121	132	174	117	162	197	208	208	251	206	166	165
	PrecMin	706	24	11	17	18	31	32	50	4	14	4	0	6
	PrecStd	165	25	30	41	26	36	42	51	45	54	46	34	44
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	15.2	19.6	25.3	28.0	31.0	32.6	34.0	30.7	25.2	20.3	14.6
		MCWB	8.6	9.7	12.1	15.6	18.6	22.1	21.0	21.5	19.8	16.2	13.1	9.3
	2%	DB	9.0	12.7	16.6	23.1	26.1	29.7	30.7	31.9	28.0	23.3	18.1	10.9
		MCWB	6.6	8.3	10.2	14.6	18.0	21.0	21.0	20.6	18.8	15.1	12.1	7.5
	5%	DB	7.0	10.9	14.3	20.9	24.2	27.8	28.6	30.0	25.8	20.8	15.9	8.8
		MCWB	5.2	7.2	9.1	13.5	17.3	20.3	20.1	20.5	17.8	14.9	11.4	6.7
	10%	DB	5.6	9.0	11.9	18.5	22.4	26.1	26.8	27.9	23.3	18.0	13.5	7.3
		MCWB	4.3	6.3	7.6	12.2	16.1	19.7	19.4	19.9	17.0	13.4	10.2	5.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	10.5	12.9	16.8	19.7	23.7	22.4	23.0	21.2	17.8	13.7	9.7
		MCDB	11.7	14.0	18.5	24.1	25.3	29.2	29.1	31.7	29.0	21.7	19.9	13.1
	2%	WB	7.0	8.7	10.9	15.3	18.8	22.2	21.7	22.0	19.6	16.3	12.7	8.5
		MCDB	8.5	11.8	15.6	21.9	24.6	27.9	28.3	29.6	25.9	21.4	17.2	10.9
	5%	WB	5.7	7.8	9.4	14.1	17.8	21.0	21.0	21.2	18.5	15.1	11.7	6.9
		MCDB	7.1	10.4	13.8	20.0	23.0	27.0	27.3	28.4	24.3	20.0	15.5	8.8
	10%	WB	4.2	6.7	8.1	12.8	16.7	20.0	20.1	20.3	17.4	13.9	10.6	5.7
		MCDB	5.5	8.9	11.5	17.6	21.4	25.5	26.1	27.0	22.4	17.3	13.5	7.5
Mean Daily Temperature Range	5% DB	MDBR	6.6	7.6	9.3	11.0	10.8	10.9	11.4	11.9	10.5	10.3	8.2	5.8
		MCDBR	8.3	10.6	12.8	14.5	14.1	13.3	14.2	15.2	14.3	14.8	11.5	8.1
		MCWBR	5.8	6.7	7.5	7.3	7.4	6.2	6.3	6.3	6.4	8.1	6.4	5.6
		MCDBR	7.8	9.2	12.0	13.4	12.8	12.4	12.6	13.9	12.2	13.5	10.7	7.7
	5% WB	MCWBR	5.6	6.1	7.2	7.0	6.9	6.3	5.9	6.2	5.9	7.5	6.1	5.4
Clear-Sky Solar Irradiance	taub		0.326	0.356	0.401	0.455	0.437	0.434	0.453	0.455	0.417	0.382	0.353	0.327
	taud		2.291	2.235	2.159	2.044	2.160	2.208	2.165	2.153	2.221	2.295	2.320	2.325
	Ebn at Noon		740	792	809	798	830	834	811	790	784	754	701	694
	Edh at Noon		78	102	127	157	145	139	143	139	119	95	76	68
All-Sky Solar Radiation	RadAvg		1.12	1.94	3.13	4.40	5.35	5.87	5.76	5.30	3.73	2.48	1.39	0.86
	RadStd		0.16	0.27	0.46	0.46	0.53	0.44	0.51	0.49	0.44	0.30	0.19	0.12